



Density Monitoring and Decision-Support System

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Panel feedback & research adaptations

Panel Comments	Solution Used
• The threshold values used to classify traffic density should be clearly defined. • The system should clearly explain how traffic is categorised as **Low, Medium, or High**. • The dashboard should show the traffic level based on a fixed and understandable rule.	• Defined clear vehicle-count thresholds for traffic density classification. • Used vehicle count from the YOLOv8 video analysis output. • Classified traffic level using the following rule: 0 vehicles → CLEAR 1 – 4 vehicles → LOW 5 – 10 vehicles → MEDIUM 11+ vehicles → HIGH

Dencity Monitoring and decision-support system

Problem & Impact

- Manual traffic monitoring is slow and difficult
- Vehicle density is not clearly measured
- Signal timing decisions are often not data-based
- Traffic congestion increases waiting time
- Officers need quick decision-support from video data

Target Users / Beneficiaries

- Traffic police officers
- Traffic control centres
- Road management authorities
- City planners
- Drivers / road users
- Public transport users



Solution & User Requirements

How the Solution Addresses the Sub-Problem

1. Officer uploads recorded traffic video
2. YOLOv8 detects and counts vehicles
3. System classifies traffic as CLEAR, LOW, MEDIUM, or HIGH
4. Queue length and clearance time are estimated
5. Suggested green signal time is given as decision support



Video Upload → Vehicle Detection → Density Classification → Queue Estimate → Green Time Suggestion

User Requirements Addressed

01 Traffic officer can upload recorded traffic videos

02 System can detect and count vehicles automatically

03 Traffic density levels are shown clearly

04 Queue length and clearance time are calculated

05 Suggested green signal duration supports officer decisions

06 Analysis records and location details are stored for review

Prototype uses uploaded video analysis; future upgrade: live CCTV / IP camera feed

Design Excellence & User Feedback

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User Feedback on Prototype

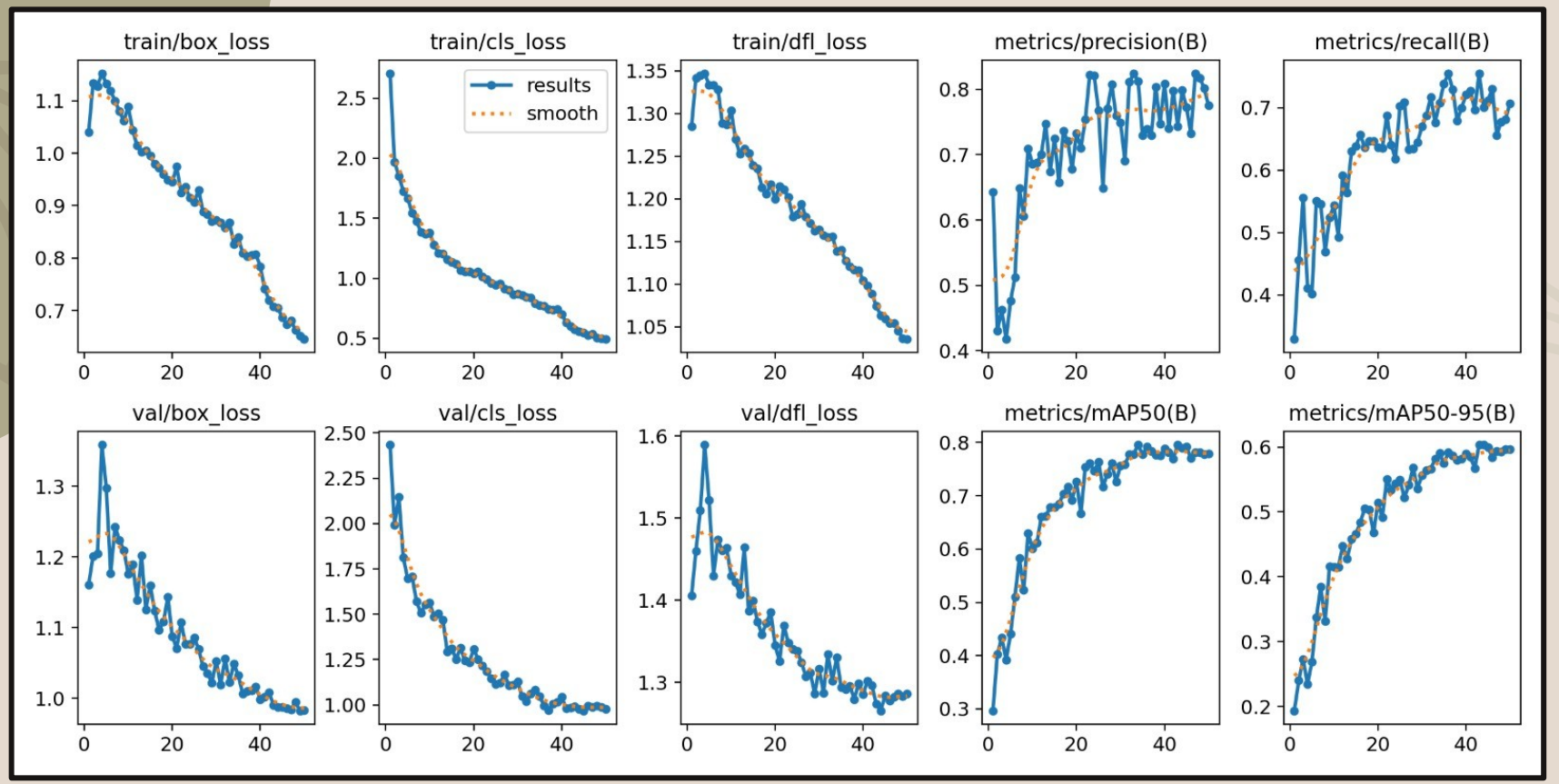
1. Traffic density threshold values should be clearly defined
2. Data source and prototype limitations should be clearly stated
3. Queue length and green time results should be easier to understand
4. The dashboard should clearly separate decision support from real signal control
5. Location and analysis records should be presented more clearly

Design Improvement Applied

Clear threshold rules, uploaded-video data source labels, queue intelligence outputs, and prototype limitation notes were added to improve usability.

Prototype uses uploaded video analysis; future upgrade: live CCTV / IP camera feed

Excellence & Validation



- The YOLOv8 training results show decreasing losses and increasing **precision, recall, and mAP values.** This proves that the model learned vehicle detection effectively .

Future Implementation



Future Implementations

Next enhancements for the Density Monitoring and Decision-Support System

1. **Live CCTV / IP Camera Integration**
   Replace uploaded videos with live traffic feeds
2. **Smart Signal Controller Integration**
   Use real signal timing instead of simulated values
3. **Multi-Junction Monitoring**
   Track congestion across multiple junctions
4. **Local Dataset Fine-Tuning**
   Train with Sri Lankan traffic videos for better accuracy
5. **Driver Awareness System**
   Show estimated waiting time and alerts for drivers
6. **Advanced Queue Estimation**
   Improve lane-wise queue and vehicle flow analysis
7. **Reports & Analytics**
   Generate peak-hour, daily, and monthly traffic reports
8. **Cloud Deployment**
   Support scalable access, storage, and remote monitoring

Current prototype uses uploaded traffic video analysis; future versions aim for live, scalable, real-time decision support.

Commercialization, Sustainability & Market Value

Section	Key Points
Commercialization	Web-based tool for traffic police, councils, and transport authorities. Can be offered as a subscription platform or custom deployment.
Sustainability	Starts with uploaded video analysis, reducing initial cost. Can later upgrade to live CCTV/IP camera integration.
Market Value	Supports vehicle counting, queue estimation, green-time suggestion, peak-hour analysis, and better traffic decision-making.

THANK YOU!